NO.6 MASTER WARNING LIGHT ILLUMINATED/MESSAGE RELATED TO VEHICLE SYSTEM MALFUNCTION INDICATED IN DISPLAY [SKYACTIV-D 1.5]

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6	MASTER WARNING LIGHT ILLUMINATED/MESSAGE RELATED TO VEHICLE SYSTEM MALFUNCTION INDICATED IN DISPLAY
DESCRIPTION	• Master warning light illuminated. • Message related to vehicle system malfunction indicated in display.
POSSIBLE CAUSE	• PCM DTC is stored. — Brake switch malfunction — Engine oil solenoid valve malfunction — Power supply system malfunction — Fuel injection amount learning not completed — Current sensor malfunction • Blow-by heater relay malfunction • Blow-by heater malfunction • Electric water pump malfunction • Water-cooled charge air cooler malfunction • speed sensor assy (turbocharger with variable turbine geometry) malfunction • Timing chain malfunction • Instrument cluster malfunction

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**VERIFY PCM DTC** • Retrieve PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**DETERMINE IF MALFUNCTION IS DUE TO ELECTRIC WATER PUMP STUCK** • Access the PID/DATA monitor item E_WP and simulation item E_WP using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the electric water pump operation sound change according to the simulation function operation?	Yes	Go to the next step.
		No	Replace the electric water pump, then go to Step 6. (See ELECTRIC WATER PUMP REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
3	**INSPECT SPEED SENSOR ASSY. (TURBOCHARGER)** • Reconnect all disconnected connectors. • Inspect the speed sensor assy. (turbocharger). (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the speed sensor assy. (turbocharger), then go to Step 6. (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT SEALING OF WATER-COOLED CHARGE AIR COOLANT PASSAGE** • Perform the ÅgWATER-COOLED CHARGE AIR COOLER COOLANT LEAKAGE INSPECTIONÅh • Does the radiator cap tester needle drop even though there is no engine coolant leakage from the radiator or the hoses?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 6.
		No	Go to the next step.
5	**INSPECT INSTRUMENT CLUSTER** • Inspect the instrument cluster. (See INSTRUMENT CLUSTER INSPECTION.) • Is the instrument cluster normal?	Yes	Go to the next step.
		No	Replace the instrument cluster, then go to the next step. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)

STEP	INSPECTION	RESULTS	ACTION
6	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		